ModelIt!

Student Activity Pack

Does More Beavers Mean a Bigger Dam?

How Beaver Builders Shape Their World

Grade NE | NGSS 3-LS4-3, 3-LS4-4, 3-5-ETS1-1 | Cause and Effect

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **Have you ever seen a beaver or a beaver dam? Where?**

2. **Why do you think beavers build dams?**

3. **Draw what you think a beaver dam looks like from the side.**

4. **What do you think happens to the water when a dam blocks a stream?**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Number of Beavers | Trees Available | Dam Size | Pond Size

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Beaver Family

Medium beavers, lots of trees

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: No Trees!

Lots of beavers, no trees

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Lone Beaver

One beaver, lots of trees

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Rainfall — How much rain falls — more rain means more water flowing toward the dam

Number of Other Animals — How many fish, frogs, and birds move into the new pond habitat

Dam Strength — How well the dam holds up over time — stronger dams last longer

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

What happens to a pond when beavers build a dam? Use evidence from your simulation to support your answers.

1. **Why do beavers need BOTH enough family members AND enough trees to build a good dam?**

2. **What would happen to all the animals living in the pond if the beavers left?**

3. **How is what beavers do similar to what human builders do?**

4. **Why do scientists call beavers 'ecosystem engineers'?**

5. **How did your model help you understand something about nature?**

8 STEM Challenge: Build-a-Dam Challenge

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YOUR ENGINEERING MISSION

Design and build a miniature dam using craft sticks and clay that holds back the most water in a plastic bin.

THE SCENARIO

A nature park wants to understand how beaver dams work. Your engineering team has been hired to build a model dam and test how much water it can hold back. Use what your model taught you about what makes a dam successful!

Engineering Design Thinking

1. WHAT SHAPE WILL YOUR DAM BE — STRAIGHT, CURVED, OR V-SHAPED?
2. HOW WILL YOU SEAL THE GAPS BETWEEN THE STICKS?
3. HOW WILL YOU MEASURE HOW MUCH WATER YOUR DAM HOLDS BACK?
4. WHAT CAN YOU CHANGE TO MAKE YOUR DAM HOLD MORE WATER?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Wildlife Biologists study how animals like beavers change their environments. They work in forests, wetlands, and national parks to protect wildlife. They earn $50,000–$85,000/year.